Digital Gold as a Frozen Compiler

Automaton, Jnana, and a Constitution-Bound Store-of-Value

Whether the resulting asset is secure, valuable, decentralized, energy-proportionate, legitimate, or competitive remains an empirical and institutional question.

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Abstract

This paper develops a constitutional-commitment architecture for a digital store-of-value. Its central proposal is that the validity and issuance law of a named asset can be expressed as a bounded semantic artifact, lowered into a deterministic verification kernel, committed at genesis, and denied any predecessor-preserving in-protocol amendment path. A changed law may found a successor or competing asset, but it does not silently retain the original constitutional identity.

The source names the proposed Proof-of-Work system **Automaton** and places it within Jnana's separation of canonical meaning, admissibility law, procedural execution, external effect, and receipt lineage. The integrated paper sharpens the claim. A frozen kernel can stabilize one technical identity relation; it cannot eliminate social choice, implementation defects, cryptographic breakage, hardware and network dependencies, market coordination, naming disputes, or political power. A content hash proves exact commitment under a declared scheme, not monetary legitimacy or semantic correctness. Independent implementations need demonstrated conformance to the same transition relation; byte-identical compilation is one configuration, not a universal requirement.

The revision also separates block proposal, ordering pressure, validation, issuance, fee formation, identity, and finality. It withdraws unsupported numerical energy estimates and replaces them with an explicit economic model. It treats MosaicID as a key-derived identifier rather than proof of uniqueness, Sybil resistance, operator identity, or software execution. It treats quorum finality as an additional trust and liveness regime whose authority, membership, rotation, collusion, and recovery rules must themselves be constitutionalized.

The resulting thesis is architectural and conditional: a frozen, reproducible, minimal verification constitution can make rule change explicit as an identity split and reduce one class of governance ambiguity. Whether the resulting asset is

secure, valuable, decentralized, energy-proportionate, legitimate, or competitive remains an empirical and institutional question.

1. Scope and Central Claim

A store-of-value is not defined by one property. Scarcity, durability, divisibility, transferability, security, liquidity, legal treatment, social recognition, and credible commitment can all matter. This paper isolates one architectural contribution: **constitutional continuity**.

The proposal does not establish investment value, financial suitability, price stability, or superiority to any existing asset. It asks a narrower systems question:

> Can the rule set that defines a digital asset be made explicit enough that any change to its validity or issuance law is mechanically recognizable as a different constitutional identity rather than a silent continuation?

The source answer is the Frozen Constitution Principle. The integrated answer preserves it with a claim ceiling.

2. Frozen Constitution Principle

2.1 Definition

Let C_0 be the genesis constitutional artifact, V_0 the admitted verification kernel, G_0 the genesis parameters and state, and S_v the exact schemas and canonicalization rules. Define a technical constitutional identity commitment:

$$I_0 = H(S_v \parallel C_0 \parallel V_0 \parallel G_0).$$

For chain identity I_0, block admission is determined by the predecessor-bound transition relation encoded by V_0:

$$\mathrm{Accept}_{I_0}(s, b) = V_0(C_0, s, b) \in \{0, 1\}.$$

The **Frozen Constitution Principle (FCP)** is:

> The state machine of I_0 contains no transition that replaces or widens C_0 or V_0 while claiming uninterrupted constitutional identity I_0.

An alternative constitution C_1 or verifier V_1 can be published and used. Under FCP it receives a distinct identity commitment I_1, unless an equivalence relation explicitly admitted by C_0 proves that the difference is representational rather than constitutional.

2.2 What is frozen

The design must state what belongs to the frozen surface:

- canonical serialization and parsing;
- cryptographic algorithms and domain separation;
- state transition and validity predicates;
- issuance and fee accounting;
- ordering and finality predicates;
- numeric ranges, overflow, time, and difficulty semantics;
- genesis state and identity derivation;
- permitted implementation equivalence;
- network messages that affect consensus;
- error and refusal behavior; and
- any validator, oracle, or attestation regime used by validity.

Leaving one of these surfaces implicit can recreate amendment through interpretation.

2.3 What FCP does not freeze

FCP does not freeze:

- which people choose to run the system;
- market price, liquidity, legal status, or cultural meaning;
- operating systems, hardware, networking, or user interfaces;
- cryptographic security against future attacks;
- the absence of undiscovered implementation defects;
- names, tickers, exchanges, or public narratives; or
- social decisions about which fork, implementation, or history receives attention.

Constitutional finality is therefore a bounded technical relation, not the end of politics.

3. Jnana and the Compilation Boundary

Jnana separates canonical semantic commitment, admissibility law, procedural lowering, domain effect, and receipt lineage [JNANA] [JNANA-II]. Applied here:

1. **MCIR** represents the bounded monetary constitution: state types, validity, issuance, authority, invariants, refusal, and nonclaims.
2. **MCIR-Law** determines whether a proposed artifact, compiler, kernel, or transition is admissible under that constitution.
3. **PEIR** is a lowered procedural form used to realize the transition relation.
4. **Domain Effects** are ledger-state and network consequences admitted through the kernel.
5. **ReceiptLog** preserves exact inputs, roots, decisions, outcomes, and predecessor relations.

The constitution is not merely prose, but neither is every executable artifact a constitution. It acquires standing only through a declared founding authority, schema, admission process, genesis commitment, and public claim boundary. The compiler is not sovereign, and the compiled verifier is not self-legitimating. Authority derives from the founding constitution and the participants who knowingly adopt its exact identity; execution enforces only the rules it actually contains.

3.1 Completeness is relative

The source calls MCIR complete with respect to validity. The stronger formulation is:

> C_0 is closed only relative to a declared consensus-relevant input and effect surface.

If parsing, cryptography, time, networking, resource exhaustion, storage, or external attestations can change validity, they are part of the surface and cannot be omitted from the completeness claim.

4. Reproducibility, Diversity, and the Trusted Computing Base

4.1 Byte identity versus conformance equivalence

Multiple compilers need not always produce byte-identical machine artifacts to implement the same transition relation. Byte identity is valuable when a canonical toolchain, target, and build environment are deliberately fixed. Diverse implementations may instead provide resilience against common-mode compiler defects.

The required claim should be explicit:

- **Reproducible-build claim:** identical declared inputs and environment produce identical bytes.
- **Conformance claim:** distinct implementations accept and reject the same defined input domain and produce equivalent successor states.
- **Semantic-equivalence claim:** two artifacts preserve the same declared constitutional meaning under a governed lens.

These claims require different evidence. The Three Hash Rule and semantic-hashing work prevent an artifact hash from standing in for constitutional or semantic equivalence [THR] [SH].

4.2 Minimal kernel, non-minimal environment

The conceptual interface is:

```
VerifyBlock(previous_state, block) -> accept | reject,
successor_state, receipt
```

The transition code can be small while the effective trusted computing base remains larger. It may include:

- parser and serialization library;
- cryptographic implementations;
- compiler and linker;
- runtime, operating system, and hardware behavior;
- storage integrity;
- network input handling;
- build and distribution custody; and
- any attestation or oracle verifier.

The architecture should minimize and independently test these dependencies rather than claiming they disappear.

4.3 Determinism contract

A consensus-critical implementation should specify canonical serialization, bounded arithmetic, overflow behavior, ordering, time, resource bounds, cryptographic domains, error outcomes, and version identifiers. Floating point is not automatically forbidden if an exact portable semantics is specified, but avoiding it may reduce the conformance surface.

5. Separating Monetary and Consensus Functions

The source identifies security, validation, and issuance as distinct. Integration yields a more exact decomposition:

Function	Question
Transaction proposal	Which candidate state changes enter the mempool or block?
Block production	Who proposes an ordered batch?
Ordering pressure	How are competing histories weighted?
Validation	Which blocks and state transitions satisfy V_0?
Issuance	How are new units created and assigned?
Fee formation	How are scarce inclusion resources priced?
Propagation	How do proposals and blocks reach peers?
Finality	When does the system treat reversal as outside the admitted history?
Identity	Which constitution and history does a node claim to follow?

In common Proof-of-Work systems, miners propose and order blocks, while independently operated full nodes also validate them. The functions are coupled by protocol and incentives, but mining does not possess exclusive validation authority. Automaton's contribution is better stated as **contractual separation and explicit coupling**, not as discovery that all existing systems literally collapse every function into one actor.

6. Proof-of-Work and Proof-of-Useful-Work

6.1 Proof-of-Work as a configured ordering mechanism

Proof-of-Work can provide a mechanically checkable cost signal and history-selection rule. It is not institution-free. Its security depends on hardware markets, energy access, network topology, mining concentration, software, economics, jurisdiction, and the assumed behavior of participants.

Automaton may choose PoW because its verification predicate is bounded and relatively independent of the semantic content of external tasks. This is a CM2 preference within the design, not proof that PoW is uniquely minimal or optimal.

6.2 Useful work is not categorically inadmissible

Proof-of-Useful-Work adds questions about task definition, result verification, data rights, benchmark change, fraud, and who receives benefit. If validity depends on mutable institutional judgments, FCP is weakened. But a useful-work predicate could remain constitutionally admissible if its task generation, verification, cost, ownership, and update rules are completely inside the frozen and mechanically checkable surface.

The defensible conclusion is conditional:

> Useful work is incompatible with this frozen-constitution profile when its validity or reward depends on unbounded, mutable, or institutionally adjudicated semantics not already committed by C_0.

AI training and inference commonly present exactly those difficult surfaces, but publication-level comparison requires current external evidence.

7. Forks, Continuity, and Asset Identity

FCP makes one identity split technically explicit. If $I_1 \neq I_0$, the artifacts are not the same constitutional commitment. This does not settle every asset-identity question.

A fork may inherit history, balances, users, brand, ticker, exchange listings, or social recognition. Market participants may call the successor the original, or reject the technical root as insufficient. The architecture can say:

- which genesis constitution a node claims;
- which transition relation a block satisfies;

- which predecessor chain a state derives from; and
- whether a rule change occurred under the configured identity scheme.

It cannot force the public to use one name or assign one economic value. Technical identity, publication identity, social identity, and market identity remain distinct.

8. Energy and Security Economics

The source gives numerical estimates for Automaton energy consumption and relative hash rate. No exact model, data, code, date, market assumptions, or sensitivity analysis is bound to those figures. They are withdrawn from the current claim set.

Let miner revenue per period be:

$$R_m = P_a(S_m + F_m),$$

where P_a is asset price, S_m is miner-directed subsidy, and F_m is fee revenue. A stylized equilibrium hash rate can be written:

$$Q^* = f(R_m, p_e, \eta_h, C_h, r, \kappa),$$

where p_e is energy price, η_h hardware efficiency, C_h capital cost, r financing and risk, and κ captures concentration, curtailment, jurisdiction, and strategic behavior.

Setting $S_m = 0$ can reduce miner revenue relative to a counterfactual with the same price and fees. It does not by itself determine a 30-50 percent ratio. Price, fee demand, hardware, difficulty, attack incentives, treasury issuance, and participant expectations can change endogenously.

An energy claim requires at least:

- exact issuance and fee model;
- transaction and settlement assumptions;
- asset-price scenarios;
- miner hardware and energy-price distributions;
- difficulty adjustment and entry/exit dynamics;
- validator, node, storage, network, oracle, and finality overhead;
- attack and recovery scenarios;
- sensitivity and uncertainty intervals; and
- reproducible code and data with dated sources.

Lower honest hash expenditure generally lowers the contemporaneous cost of acquiring a competing hash majority, all else equal. Any compensating finality mechanism introduces its own trust and failure assumptions.

9. Bitcoin Comparison as a Research Program

The source argues that soft forks, dominant clients, economic coalitions, and changing fee economics leave interpretive surfaces in Bitcoin. These are plausible comparison hypotheses, not established facts within the evidence available to this pass.

For a future evidence-qualified revision, each comparison should be rebuilt from current primary sources and exact cases:

1. Which changes count as consensus rule tightening, implementation correction, policy change, or chain split?
2. Which actors possessed proposal, implementation, signaling, adoption, and enforcement power?
3. What did non-upgraded validating nodes accept or reject?
4. Did a claimed semantic change preserve, narrow, or replace the original validity set?
5. How were branding, economic majority, and technical lineage related?
6. What counterexamples show successful resistance to concentrated influence?

The paper's internal contribution is the analytical lens: mutation can be distinguished from continuity by binding a technical identity to exact constitutional roots and an explicit equivalence rule. It does not establish that Bitcoin is captured, that one implementation is sovereign, or that Automaton eliminates economic power.

10. MosaicID: What a Key-Derived Identifier Can and Cannot Prove

Suppose a MosaicID is defined as:

$$\text{MID} = H(\text{public_key}).$$

This can supply a stable identifier for a cryptographic key. A signed declaration can bind that key to a claim about constitution I_0. It can support authentication, message attribution to the key, equivocation evidence, and revocation or succession rules if those rules are specified [MIP].

It does **not** by itself prove:

- one key per person, organization, machine, or miner;
- resistance to creating many keys;
- the legal or social identity of the operator;
- that the operator controls no other keys;
- that software actually executes V_0;
- that a message was the first network broadcast;
- that a pool label identifies all contributing miners; or
- that the key remains uncompromised.

Peer-layer Sybil resistance therefore requires scarce capabilities, admission policy, resource tests, topology controls, rate limits, reputational cost, or another mechanism beyond a cheap key hash. Execution claims require measured boot, reproducible attestation, challenge-response, or behavioral evidence, each with its own trusted base.

A genesis artifact published in a sealed DKN dimension can preserve custody if the dimension, publication time, membership, key destruction, replication, and verification rules are independently evidenced. Key burning is not self-proving, and a signed declaration does not establish that a constitution predates all later alternatives without a trusted time or ordering relation.

11. Quorum Finality as an Additional Constitution

The source proposes MosaicID-signed quorum attestations from a founding validator set. This is not merely a remedy added outside FCP. If quorum determines validity or irreversibility, its membership, threshold, message schema, equivocation rule, expiry, rotation, recovery, and failure semantics belong inside C_0.

Let W_0 be the founding validator set, w_i a declared weight, and θ the threshold. A configured finality predicate may be:

$$\mathrm{Final}_{I_0}(b,h) = 1 \quad \text{iff} \quad \sum_{i \in A(b,h)} w_i \geq \theta$$

and every attestation in $A(b,h)$ is valid, non-expired, non-equivocating under the exact schema, and bound to I_0, block b, and height h.

This establishes finality only for clients whose constitution recognizes that predicate. It does not make the block metaphysically irreversible. A later implementation can ignore the quorum rule, but under FCP it is a different constitutional identity.

11.1 New failure surfaces

Quorum finality adds:

- validator capture or key compromise;
- censorship and selective non-attestation;
- liveness failure when threshold cannot be reached;
- correlated infrastructure and jurisdiction;
- founding-set entrenchment;
- unsafe key loss;
- equivocation detection without effective consequence;
- disagreement about time, height, or eligible membership;
- validator-miner ownership overlap; and
- pressure for emergency rotation or recovery that conflicts with freezing.

Hash majority, validator compromise, and timing are not automatically independent. The same capital, infrastructure, providers, jurisdictions, or actors may influence several. Security claims need an explicit dependency graph and attack-cost model rather than multiplying nominal surfaces.

11.2 Exclusion is governance

Removing federation access after equivocation is an externally consequential sanction. The system must specify evidence, adjudication, false-positive handling, appeal, proportionality, key succession, and the exact authority that enforces exclusion. Calling the consequence protocol-level does not remove its governance character.

12. Automaton Identity Across the Corpus

The source paper uses **Automaton** for a PoW digital-gold design with a frozen verifier. A later *Automaton Constitutional Assets Domain Codex* expands the name to a three-currency monetary semantic reactor containing Sovereigns, Governors, Commons, and three frozen functions [ACA-CODEX].

This paper should therefore be read provisionally as one of the following:

1. the original Automaton architecture;
2. the Sovereigns or store-of-value component of the later ACA architecture;
3. a predecessor work whose name was broadened by a successor codex; or
4. a distinct frozen-compiler proposal sharing the Automaton label.

The corpus contains evidence for evolution but not an author-adjudicated identity merger. This revision does not decide among the four. Any public version should state the relation explicitly.

13. Corpus Integration

13.1 Jnana and MCIR

Jnana supplies the semantic-to-procedural layering and insists that transformation legitimacy be governed alongside execution [JNANA] [JNANA-II]. The frozen-compiler paper is a domain profile of that stack, not evidence that Automaton implements every Jnana layer.

13.2 Three Hash Rule and Semantic Hashing

The Three Hash Rule separates object state, admissibility, and meaning commitments [THR]. Semantic Hashing adds a declared equivalence lens [SH]. Automaton should therefore avoid using one genesis hash to claim exact state, constitutional admissibility, and semantic continuity simultaneously.

13.3 Admissibility and constraint carriage

Admissibility Engineering supplies `ADMIT`, `DEFER`, `REFUSE`,
and `GAP` rather than a binary claim that every environmental condition is reducible to `accept|reject` [AE]. Constraint-Bearing Composability requires constitutional obligations to survive compilers, clients, network messages, receipts, and finality interfaces [CBC].

13.4 Receipts and work custody

Receipts bind exact inputs, decision rules, outcomes, and claim ceilings [RBC] [FTC]. They do not make a false decision true. Build, compiler, artifact, genesis, block, attestation, and exclusion custody should remain distinct.

13.5 MIP and DKN

MIP supplies candidate identity and network-governance mechanisms [MIP]. DKN can preserve genesis and evidence artifacts. Neither lexical connection proves that a particular MIP or DKN implementation realizes Automaton, and custody-only excluded systems are not semantic evidence.

14. Realization Funnel

Stage	Frozen-compiler work
Possibility	Explore monetary constitutions, consensus families, amendment models, no-update forms, useful-work predicates, identity rules, failure recovery, harmful lock-in, and plural asset forks.
Specification	Define state, validity, issuance, serialization, cryptography, ordering, identity, finality, authority, evidence, privacy, nonclaims, and failure semantics.
CM1	Enumerate admissible configurations, including insecure, centralized, energy-intensive, illiquid, surveillance-heavy, brittle, captured, or unusable forms.
CM2	Prefer configurations with minimal authority, exact identity splits, bounded semantics, diverse verification, recoverable custody, explicit dependencies, and evidence proportional to risk.

Stage	Frozen-compiler work
Instance Configuration	Bind exact MCIR, compiler and toolchain, kernel, genesis, algorithms, parameters, network, validator set, thresholds, clients, hardware assumptions, economic model, and courts.
Implementation Realization	Build and independently implement the kernel, node, compiler, custody, receipts, network, identity, finality, recovery, and user interfaces.
Evidence	Execute differential, adversarial, economic, cryptographic, networking, liveness, safety, fork, custody, reproducibility, privacy, and cold-boot tests.
Revision	Publish successor analyses, implementations, or assets with explicit new roots; never present a changed constitution as the old exact identity.

The source provides a domain proposal and portions of Specification. It does not establish an operational currency, evidence-qualified implementation, or financial claim.

15. Evidence and Conformance Program

A bounded frozen-compiler conformance claim should require:

1. exact roots for source constitution, schema, canonicalization, compiler, dependencies, kernel, genesis, tests, and receipts;
2. a closed enumeration of consensus-relevant input and effect surfaces;
3. formal or executable state-transition semantics with total error behavior;
4. reproducible-build evidence for any byte-identity claim;
5. multiple independent implementations and differential testing for any conformance-equivalence claim;
6. adversarial serialization, arithmetic, cryptography, resource, storage, and network fixtures;
7. proof that no admitted state transition changes the frozen constitutional roots;
8. explicit fork and identity-split tests;
9. chain safety and liveness under declared network and adversary models;
10. economic simulations with code, data, dates, sensitivity, and uncertainty;

11. MosaicID tests that distinguish key authentication from uniqueness, Sybil resistance, execution attestation, and human identity;
12. quorum safety, liveness, equivocation, key-loss, collusion, censorship, and correlation tests;
13. custody and replay of genesis publication, DKN storage, build, block, attestation, and sanction receipts;
14. privacy, surveillance, contest, exit, and fork analysis;
15. operational recovery without silent constitutional amendment;
16. current primary-source comparison for Bitcoin, PoW, PoUW, finality, energy, and other systems;
17. a fresh-environment cold boot; and
18. explicit nonclaims about value, investment suitability, decentralization, legitimacy, superiority, and universal security.

Constitution-Bound Monetary Automaton Thesis

Across the completed P1 corpus, Digital Gold as a Frozen Compiler is a **constitution-bound monetary automaton thesis**: selected monetary and ordering rules are compiled into a deliberately narrow, reproducible transition system whose identity and amendment boundaries are explicit. It is not a demonstrated currency, an empirical superiority claim over Bitcoin, or proof that code can eliminate governance.

- **Adjacent work:** *Jnana*
  Supplies: The wider constitutional compilation and execution architecture.
  Digital-gold boundary: The monetary automaton is one proposed domain realization, not Jnana itself.

- **Adjacent work:** *Everything Is a Compiler*
  Supplies: The general account of systems as transformations under implicit or explicit rules.
  Digital-gold boundary: The frozen compiler is a deliberately constrained application of that lens.

- **Adjacent work:** *Identity Spine Doctrine*
  Supplies: Separation of artifact, work, version, semantic, system, and deployment identities.
  Digital-gold boundary: Asset and automaton continuity require explicit identity rules beyond matching code or keys.

- **Adjacent work:** *Governance as Compilation*
  Supplies: Governed translation, admission, lowering, and remainder.
  Digital-gold boundary: Freezing narrows post-admission discretion; it does not remove the governance of specification and amendment.

- **Adjacent work:** *Sovereign Software*
  Supplies: Inspectability, reconstruction, delegation, provider removal, and

effective exit.

Digital-gold boundary: A monetary system's reproducibility supports but does not establish user or community sovereignty.

A monetary automaton claim should bind:

$$M = (constitution, compiler, transition_system, state, \\ ordering, issuance, finality, identity, amendment,. \\ evidence, threat_model)$$

"Frozen" must identify which rules, compiler stages, parameters, and authority surfaces are immutable and which remain configurable or amendable. Source reproducibility, byte-identical builds, transition conformance, consensus agreement, economic security, asset continuity, and social acceptance are different claims.

Forks expose this distinction. Two successors may preserve code while diverging in history, authority, community recognition, liquidity, or claims. A key-derived identifier establishes only the identity relation specified by its scheme. Quorum finality adds a governance constitution with exclusion, capture, availability, and correlated-control risks.

The universal GSM can type `GOVERNED_BY_MONETARY_CONSTITUTION`, `COMPILED_AS_AUTOMATON`, `ORDERS_TRANSITIONS_BY`, `ISSUES_UNDER`, `FINALIZES_UNDER`, `IDENTIFIES_AS_ASSET`, `FORKS_FROM_HISTORY`, `AMENDABLE_UNDER`, `REPRODUCED_BY_BUILD`, `CONFORMS_ON_TRANSITIONS`, and `SUPPORTED_BY_ECONOMIC_EVIDENCE`. Each claim remains separately testable.

This integration establishes four editorial commitments: adopt constitution-bound monetary automaton thesis as the official corpus role; adopt the monetary-claim envelope; publish a precise frozen-versus-amendable surface; and separate reproduction, transition, consensus, economic, identity, and social-standing evidence.

Domain-Specific Possibility System and Distributed-Topology Boundary

The fixed-seed comparison with a reflective-context profile, the Classified Earthworks brief, Dynatology, and CCOO separates a monetary automaton from superficial neighboring analogies:

$$M^* = (constitution, state, transitions, admission, ordering, \\ finality, identity, participants, topology, amendment,. \\ evidence, threat_model)$$

Context-compiled reflective behavior and physical earthworks routes have no direct evidentiary relation to monetary validity, security, asset continuity, or social acceptance. Their shared use of compilation or realization vocabulary is not a basis for integration beyond the general GSM.

Dynatology can model the automaton as one domain-specific possibility and transition system, provided its states, admissibility predicate, realization map, observations, and viability criteria are explicitly bound. The mapping contributes a grammar; it does not prove economic security or ontological primacy.

CCOO offers a separate experimental architecture for local constitutional participants and optional aggregation. It may provide comparison questions about topology, local operators, concentrated coordination, and displaced function. It is not the monetary protocol, and its toy results do not support monetary or consensus claims.

The universal GSM can type
`MODELED_AS_DOMAIN_DYNATOLOGICAL_SYSTEM`,
`DISTRIBUTED_ACROSS_MONETARY_PARTICIPANTS`, `COMPARED_WITH_OPTIONAL_AGGREGATION_TOPOLOGY`,
`PRESERVES_ASSET_IDENTITY_RULE`, `WITHHOLDS_ECONOMIC_STANDING`,
and
`HAS_NO_MATERIAL_RELATION_TO_REFLECTIVE_OR_`
`EARTHWORKS_EVIDENCE`.

This randomized review establishes four editorial commitments: adopt the expanded monetary-system object; use Dynatology only as a typed domain grammar; compare CCOO topology without identity merger or evidence transfer; and retain explicit nonrelations to reflective-context and earthworks claims.

A Bounded Monetary-Automaton Thesis

Only two downstream papers use this work directly, and both use it as a disciplined comparison rather than a monetary foundation: the Classified Earthworks brief tests physical analogy, and Context-Compiled Consciousness Mode tests reflective-context analogy. This limited dependency surface supports a bounded research thesis about a constitution-bound monetary automaton. It does not support claims of economic superiority, Bitcoin equivalence, asset legitimacy, deployment, or universal compiler ontology.

$$\begin{aligned} A^* = (&constitution, version, state, transition,\\ &implementation_set, consensus, ordering, finality,\\ &issuance, identity, fork, security_economics,\\ &dependencies, evidence, governance_remainder) \end{aligned}.$$

The frozen object must be named. Rules can be frozen while implementations, networks, hardware, cryptography, social recognition, markets, energy costs, or emergency practice change. Byte identity, deterministic execution, conformance

equivalence, consensus agreement, economic finality, and asset continuity are different claims. A key-derived identifier proves only the declared derivation under its cryptographic assumptions; it does not establish a person, ownership, legitimacy, uniqueness beyond scope, or continuing control.

Forks expose the constitutional remainder: formal continuity, social adoption, exchange recognition, shared history, and asset identity may diverge. Proof-of-work, useful work, quorum finality, and other ordering mechanisms require separate incentive, capture, exclusion, liveness, and security analysis.

The universal GSM can type
`BINDS_MONETARY_CONSTITUTION`,
`IMPLEMENTS_FROZEN_TRANSITION_RULE`, `CONFORMS_WITHOUT_BYTE_IDENTITY`, `ORDERS_MONETARY_EVENT`, `FORKS_ASSET_HISTORY`, `DISPUTES_ASSET_CONTINUITY`, and
`RETAINS_MONETARY_GOVERNANCE_REMAINDER`.

This review establishes four editorial commitments: retain bounded monetary-automaton standing; adopt the exact automaton record; separate rule, implementation, consensus, finality, identity, and legitimacy; and require explicit fork, energy, exclusion, capture, and governance-remainder analysis.

16. Failure Modes and Counterevidence

The proposal is weakened or fails when:

- an omitted environmental behavior can change validity;
- the frozen kernel contains an exploitable defect with no safe recovery path;
- one compiler and implementation create common-mode failure;
- byte reproducibility is confused with semantic correctness;
- users cannot tell which constitution or history they are running;
- brand and market coordination override the claimed technical identity;
- no-update design prevents repair of cryptographic or safety failures;
- PoW becomes concentrated, censorable, or economically insecure;
- fee-only security cannot fund the desired attack resistance;
- useful external work can be mechanically verified after all, weakening the categorical PoUW objection;
- MosaicID creates unlimited cheap identities or falsely implies software attestation;
- the validator quorum centralizes power, loses keys, stalls, colludes, or shares dependencies with miners;
- sanction and exclusion rules deny due process or create coercive governance;
- DKN custody, key destruction, or publication ordering cannot be independently proven;
- the later three-currency Automaton architecture contradicts the original no-dependency thesis through oracle or bridge governance;
- energy or security benefits disappear under a complete economic model; or

- a simpler genesis-committed state machine provides the same assurance without the additional terminology.

The strongest counterexample is catastrophic immutability: a defect or cryptographic break makes the original kernel unsafe, yet the architecture's defining property forbids in-place repair. FCP can preserve exact identity only by accepting abandonment, containment above the kernel, or explicit successor identity. That is a real cost, not an implementation detail.

17. Authorial Compression

This paper's novel contribution should be stated as a **monetary frozen-constitution profile of Jnana and constitutional computing**, not as proof that every named supporting system is newly foundational.

- MCIR and PEIR come from Jnana.
- exact object, admissibility, and meaning commitments come from the Three Hash Rule and Semantic Hashing;
- no-silent transition and explicit refusal come from Admissibility Engineering;
- cross-boundary obligation preservation comes from Constraint-Bearing Composability;
- receipts and lineage come from receipt-bound capabilities and work custody;
- MosaicID, MIP, DKN, and quorum finality are optional realization candidates with independent proof burdens; and
- Automaton is the domain system or profile that composes them for a monetary constitution.

The phrase **frozen compiler** remains useful because it names the decisive configuration: the constitutional transition relation is executable, genesis-bound, and denied a predecessor-preserving amendment path. The compiler is a construction boundary; the frozen verifier and identity rule are the runtime commitment.

Frozen Rules Stabilize an Identity Claim, Not Monetary Legitimacy

Public orientation

Digital Gold as a Frozen Compiler proposes that a named digital asset commit at genesis to an exact validity and issuance constitution. A rule change would create a successor or competitor rather than silently preserve the original technical constitutional identity. This can make one class of rule change explicit. It cannot freeze social meaning, eliminate governance, secure an implementation, or create value.

Adversarial tests

1. **Hash-legitimacy attack.** Exact commitment to bytes is treated as proof that the constitution is correct, fair, lawful, intelligible, or economically sound.
2. **Frozen-defect attack.** A bug, ambiguity, cryptographic weakness, unsafe parameter, or exploit is permanently protected as identity rather than safely containable.
3. **Environment-closure attack.** The kernel is frozen while compilers, hardware, networks, cryptography, time, dependencies, clients, and operational practices change its effective behavior.
4. **Fork-naming attack.** Technical lineage alone determines which community, market, custodian, exchange, court, or user may retain the original asset name.
5. **Conformance-divergence attack.** Independent implementations pass convenient vectors but diverge on serialization, arithmetic, resource limits, error handling, ordering, or adversarial states.
6. **Consensus-finality attack.** Proof of Work, quorum signatures, economic weight, or longest-chain selection is treated as truth, legitimacy, irreversibility, or protection against coordinated capture.
7. **Useful-work attack.** Claimed useful computation is unverifiable, gameable, market-distorting, or weakly coupled to the security objective.
8. **Investment-promotion attack.** Architectural language implies scarcity, safety, decentralization, sustainability, price support, financial suitability, or superiority to existing assets.

Claim and evidence ceiling

The paper offers a conditional technical identity architecture and research program. It does not establish a working network, asset value, investment suitability, security, decentralization, liveness, finality, energy proportionality, legal treatment, Sybil resistance, operator identity, software execution, novelty, or superiority to Bitcoin or any other system. Automaton remains a proposal.

Evaluation requires an exact semantics and threat model, multiple conforming implementations, adversarial state and fork tests, recovery analysis, cryptographic agility, network and incentive simulation, energy and economic measurement, legal and institutional review, and current prior-art research. No implementation, issuance, financial action, or external research occurred in this pass.

Reader-facing consequence

The defensible promise is narrow: everyone can know when the validity law changed enough that the old technical identity no longer applies. Whether anyone should trust, use, value, regulate, or preserve the resulting asset remains a separate question.

18. Research Agenda and Open Questions

1. Confirm whether constitutional finality means no in-protocol predecessor-preserving amendment, or a stronger claim.
2. Define exactly what is frozen and which environmental behaviors are consensus relevant.
3. Choose byte-identical reproducibility, conformance-equivalent diverse implementations, or both, and state their evidence contracts.
4. Decide whether PoW is required by Automaton or one CM2 configuration among several.
5. Withdraw, replace, or substantiate every energy, hash-rate, security-cost, and Bitcoin comparison with reproducible current evidence.
6. Define the relationship between the original Automaton PoW proposal, Automaton Sovereigns, and the later three-currency Automaton Constitutional Assets architecture.
7. Decide whether MosaicID is only a key identifier or also part of a separately specified admission, Sybil, attestation, and operator-identity regime.
8. Define quorum membership, threshold, rotation, key loss, liveness, equivocation, sanction, appeal, expiry, and recovery inside the frozen constitution.
9. Define the acceptable response to catastrophic bugs and cryptographic breakage: abandonment, upper-layer containment, or explicit successor asset.
10. Select any implementation or live economic test only under separate implementation and effect authority.
11. Authorize external primary-source and quantitative research for a future evidence-qualified revision readiness.

19. Conclusion

Digital gold as a frozen compiler is strongest when read as a disciplined identity and transition claim, not as a promise that politics, economics, or implementation risk disappear.

A bounded MCIR can state the monetary law. A governed compiler can lower it. A minimal verifier can enforce one exact transition relation. Genesis commitments can bind the constitution, kernel, and initial state. A no-amendment rule can ensure that a changed law receives a changed technical identity. Jnana, semantic hashing, admissibility, constraint carriage, receipts, and custody make each boundary inspectable.

The same discipline exposes the costs. A frozen defect stays frozen. A key hash is not a person, a scarce identity, or proof of execution. A founding quorum is a constitutional authority with capture and liveness risks. Lower miner revenue can lower hash security. Social actors remain free to coordinate around a successor, reuse a name, or abandon the system. Market value is not compiled.

The defensible thesis is therefore:

> A genesis-bound executable constitution can make monetary rule mutation mechanically explicit as an identity split, provided the complete validity surface, implementation equivalence, supporting authority, and evidence contracts are themselves declared and tested.

That is more precise than saying governance vanishes. Governance is moved into founding specification, explicit adoption, bounded operation, evidence, and successor choice. The compiler freezes a constitution; it does not freeze the world.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Instance Configuration.** The present paper does not claim a complete instance configuration treatment. That stage remains successor work and cannot be inferred from the architecture described here.
- **Possibility Space.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Revision.** The present paper does not claim a complete revision treatment. That stage remains successor work and cannot be inferred from the architecture described here.

Corpus-Wide Reference Positioning

This paper argues that a constitutional-commitment architecture for a digital store-of-value. Its nearest external comparison fields are crypto, formal, governance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A valid hash, signature, or inclusion proof establishes a bounded cryptographic relation, not meaning, authority, safety, or jurisdiction by itself. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Anders Rundgren, Bret Jordan, and Samuel Erdtman. *JSON Canonicalization Scheme (JCS)*. RFC 8785, 2020. https://www.rfc-editor.org/rfc/rfc8785 **Relationship:** Adjacent canonicalization standard relevant to stable cryptographic commitments.

[**E2**] Ben Laurie et al. *Certificate Transparency Version 2.0*. RFC 9162, 2021. https://www.rfc-editor.org/rfc/rfc9162 **Relationship:** Adjacent append-only Merkle-log standard for inclusion and consistency evidence.

[**E3**] Manu Sporny et al., eds. *Decentralized Identifiers (DIDs) v1.0*. W3C Recommendation, 19 July 2022. https://www.w3.org/TR/did-core/ **Relationship:** Adjacent decentralized identity standard; identifier resolution is not assumed to settle semantic standing or jurisdiction.

[**E4**] Juan Benet. "IPFS - Content Addressed, Versioned, P2P File System." 2014. arXiv:1407.3561. https://arxiv.org/abs/1407.3561 **Relationship:** Predecessor content-addressed network architecture; content identity is compared with, not treated as sufficient for, semantic jurisdiction.

[**E5**] George C. Necula. "Proof-Carrying Code." *POPL '97*, 1997, pp. 106-119. DOI: 10.1145/263699.263712. https://doi.org/10.1145/263699.263712 **Relationship:** Predecessor for binding machine-checkable evidence to executable

artifacts; the present paper may demand broader semantic or constitutional obligations.

[**E6**] Xavier Leroy. "Formal Verification of a Realistic Compiler." *Communications of the ACM* 52(7), 2009, pp. 107-115. DOI: 10.1145/1538788.1538814. https://doi.org/10.1145/1538788.1538814 **Relationship:** Predecessor verified-compilation result used to distinguish proven translation preservation from broader governance claims.

[**E7**] Patrick Cousot and Radhia Cousot. "Abstract Interpretation: A Unified Lattice Model for Static Analysis of Programs by Construction or Approximation of Fixpoints." *POPL '77*, 1977. DOI: 10.1145/512950.512973. https://doi.org/10.1145/512950.512973 **Relationship:** Predecessor formal framework for sound approximation and fixpoint reasoning.

Ableman corpus

[**A1**] Adam Ableman. *The Three Hash Rule.* Ableman Corpus current edition; work identity `PAPER.PUB.20788617`;
source SHA-256
`F6FA73D083B70711A8C817F7779751D1506FEBA0E356DB9805BB1570663731B5`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Semantic Hashing: Governed Commitment Schemes for Semantic Provenance, Admissibility, and Constitutional Continuity.* Ableman Corpus current edition; work identity `PAPER.PUB.20248734`;
source SHA-256
`B1DAC3E2CD4D57BAFBF6462BA6ACFFF7CF120CDB7FA37D0C173525744CC0A2F0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Cryptographic Admissibility and Jurisdiction.* Ableman Corpus current edition; work identity `PAPER.PUB.20602448`;
source SHA-256
`6094338D8706F63E8787A5E8DCF40F8A651C0FC3E5EA776DD61791566D094D38`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Self-Certifying Proof-Carrying Identity for Registrar-Free Namespace Networks.* Ableman Corpus current edition; work identity `PAPER.PUB.20114888`;
source SHA-256
`1AFDBD390D3BF28224727FE150FFB30CA1810D997778CC8702B6CD20C4FA5B45`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Semantic Hashing: Hashes as Declared Equivalence Classes of Meaning.* Ableman Corpus current edition; work identity

PAPER.PUB.18500533;
source SHA-256
7DA63A76DDF492FFC289F486923FB430879DC148CF3BB16EB0E755E2D4AB9C14.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *MCIR: A Canonical Intermediate Representation for Governed Transformation*. Ableman Corpus current edition; work identity PAPER.PUB.21095303;
source SHA-256
EA129ECBA95F7B07CAA7B33A54F91F52913C94A14DA227EB9A06CFD083E4E5BF.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Governed Semantic Compilation*. Ableman Corpus current edition; work identity PAPER.PUB.18453681;
source SHA-256
5FF6C636550B04DB0E2F6F94C379BB75AA0BD8781E19B047D7D3BB6D49071AC2.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Ableman Corpus references [**DGFC-SOURCE**] Adam Ableman. *Digital Gold as a Frozen Compiler*. Exact author source SHA-256
7177232C1F8EC845A818A7408F1169EB8DF8CA355DF2E408D1FEF1F614B580E9;
unpublished source revised here as an AI-assisted integrated edition.

[**DGFC-CODEX-COPY**] Adam Ableman. *Digital Gold as a Frozen Compiler v1.1*. Exact canonical-codex artifact SHA-256
7D33E96C953E821E3CD359ABF1388D7B92EE7DA6F0BC3B5D6C2424A39821112A;
retained as a distinct exact artifact without automatic version-family effect.

[**ACA-CODEX**] Adam Ableman. *Automaton Constitutional Assets Domain Codex v1.0*. Exact source SHA-256
86B75825F3C706A93E49832191C7E89A532A12237420CA0FC30F3A976573C5D3;
later architecture evidence, not automatic identity adjudication.

[**JNANA**] Adam Ableman. *Jnana*. Exact reviewed source SHA-256
B5238D73BA03320D0D331A3D40AEEAB4264F41EA0DF50DA0AF61F769307F109D;
separately custodied reviewed source.

[**JNANA-II**] Adam Ableman. *Jnana II: Constitutional Transformation Substrate*. Exact reviewed source SHA-256
7F66DCD868229A4DFED7E91DD5A4647A1C54159B77128F464BFF5A9962634AF0;
separately custodied reviewed source.

[**THR**] Adam Ableman. 2026. *The Three Hash Rule*. Zenodo. DOI:
10.5281/zenodo.20788617. Exact reviewed source SHA-256
03F60317CC9AFB36DDBCF9BF1C6D185611666CC4D9AB290BD4BCC17DF4579B7C.

[**SH**] Adam Ableman. 2026. *Semantic Hashing: Hashes as Declared Equivalence Classes of Meaning*. Zenodo. DOI: 10.5281/zenodo.18500533. Exact reviewed source SHA-256 0AD135E1A89CE3EE136F36C703C3A415E1209B02FB16A59182F8E29F69127061.

[**AE**] Adam Ableman. *Admissibility Engineering*. Exact reviewed source SHA-256 FA80D5895A569839958193735B754BE62EE20E1B399A06CB3BA959614DEBF7A3; separately custodied reviewed source.

[**CBC**] Adam Ableman. *Constraint-Bearing Composability*. Exact reviewed source SHA-256 BEEA0B56DBF7DC5274307E95E08AF0576A9F91A729C1962765361ADA34488B62; supplied Zenodo title match, with exact publication identity retained separately.

[**RBC**] Adam Ableman. 2026. *Receipt-Bound Capabilities: Membrane Tokens for Governed Interaction with Reality*. Zenodo. DOI: 10.5281/zenodo.18601631. Exact reviewed source SHA-256 404EE1CCAFB63359190E45846852B8584C915EC9E76D55140024C537359D815C.

[**FTC**] Adam Ableman. *A Formal Theory of Work Custody*. Exact reviewed source SHA-256 49C38F60B95E107F8B3D864ACAFB4C49067A146C807E5248307DE7B009AD4696; separately custodied reviewed source.

[**MIP**] Adam Ableman. *Mosaic Internet Protocol*. Exact reviewed source SHA-256 0E8274185D0DACD0D0F01FE292FA854EBAAFDDEEC8F8AF70AF3194F3ED3C05DB; supplied Zenodo title match, with exact publication identity retained separately.

[**CC**] Adam Ableman. *Constitutional Computing*. Exact reviewed source SHA-256 A1EF0D1F7B5DF84A107E986421EAA95827AEDCB801B7DA63F72BEECA6047BE8D; supplied Zenodo title match, with exact publication identity retained separately.

No external research was conducted in this pass. All claims about current cryptocurrency protocols, energy use, market behavior, security economics, and comparative novelty require current primary-source verification for a future evidence-qualified revision readiness.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus

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DOI: 10.5281/zenodo.23847633